

MATH 1564 — Notes 09/03

A function f is a device with input x and output $f(x)$.

One function is $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$ ($\mathbb{R}^n$ is the **domain**, $\mathbb{R}^m$ is the **codomain**).

Example: What's the domain and codomain of x^2 ? Domain $\in \mathbb{R}$, codomain $\in \mathbb{R}$.

Definition: The image (the range) of f is the set $\{f(x) : x \text{ is in the domain}\}$.

Example: The image of $f(x) = x^2$ is $f(x) \geq 0$.

Example: Domain, codomain, and image of $f(x) = \frac{1}{x}$?

$$\text{Domain} = \mathbb{R} - \{0\}, \quad \text{Codomain} = \mathbb{R}, \quad \text{Image} = \mathbb{R} - \{0\}.$$

Example: Let A be an $m \times n$ matrix. Define a function $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ given by $T(x) = Ax$.

$$\text{Domain of } T = \mathbb{R}^n, \quad \text{Codomain of } T = \mathbb{R}^m.$$

Theorem 1: If $A = [a_1 \ \cdots \ a_n]$, then the image of T is $\text{span}\{a_1, a_2, \dots, a_n\}$.

Proof: Since a_i = column i of A ,

$$T(x) = [a_1 \ \cdots \ a_n] \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} = x_1 a_1 + \cdots + x_n a_n.$$

Since any x is allowed, this covers all linear combinations of $a_1, \dots, a_n$.

Example: Let A be the zero matrix. Then $T(x) = Ax = 0$ for all x .

Example: $T : \mathbb{R}^n \rightarrow \mathbb{R}^n$ and $A = \begin{bmatrix} 1 & & 0 \\ & \ddots & \\ 0 & & 1 \end{bmatrix}$, the $n \times n$ identity matrix.

$$T(x) = \begin{bmatrix} 1 & & 0 \\ & \ddots & \\ 0 & & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} = x_1 \begin{bmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix} + x_2 \begin{bmatrix} 0 \\ 1 \\ \vdots \\ 0 \end{bmatrix} + \cdots + x_n \begin{bmatrix} 0 \\ \vdots \\ 0 \\ 1 \end{bmatrix} = x.$$

Example: $T : \mathbb{R}^n \rightarrow \mathbb{R}^n$ and $A = \begin{bmatrix} c & & 0 \\ & \ddots & \\ 0 & & c \end{bmatrix}$ for some scalar c . Then
 $T(x) = cx$.

Example: Projection of $\mathbb{R}^2$ onto the x_1 -axis:

$$T\left(\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}\right) = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} x_1 \\ 0 \end{bmatrix}.$$

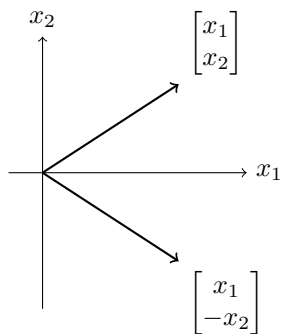
Example: Projection of $\mathbb{R}^2$ onto the x_2 -axis:

$$T\left(\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}\right) = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ x_2 \end{bmatrix}.$$

Reflection across the x_1 -axis.

Example: $A = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$,

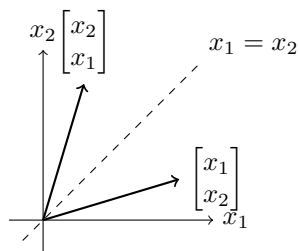
$$T\left(\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}\right) = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} x_1 \\ -x_2 \end{bmatrix}.$$



Reflection across $\{x_1 = x_2\}$.

Example: $A = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$,

$$T\left(\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}\right) = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} x_2 \\ x_1 \end{bmatrix}.$$



Definition: A function (map, or transformation $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$) is **linear** if:

a) $T(x + y) = T(x) + T(y) \quad \forall x, y \in \mathbb{R}^n$

b) $T(cx) = cT(x) \quad \forall c \in \mathbb{R}$

Example (proof): $T(x) = Ax$ is linear.

$$T(x + y) = A(x + y) = [a_1 \ \cdots \ a_n] \begin{bmatrix} x_1 + y_1 \\ \vdots \\ x_n + y_n \end{bmatrix} = (x_1 + y_1)a_1 + \cdots + (x_n + y_n)a_n$$

$$= x_1a_1 + \cdots + x_na_n + y_1a_1 + \cdots + y_na_n = Ax + Ay = T(x) + T(y).$$

Similarly, $T(cx) = A(cx) = cAx = cT(x)$.

Theorem 2: Let $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ be a map. Then T is linear $\iff$

$$T(c_1v_1 + \cdots + c_kv_k) = c_1T(v_1) + \cdots + c_kT(v_k)$$

for all vectors v_i and scalars c_i , $i \in [1, k]$.

Proof: Take $k = 2$ and $c_1 = 1 = c_2$, and $k = 1$ (these are exactly properties (a) and (b) of linearity). Induct on k :

$$\begin{aligned} T(c_1v_1 + \cdots + c_kv_k) &= T(c_1v_1 + \cdots + c_{k-1}v_{k-1}) + T(c_kv_k) \\ &= c_1T(v_1) + \cdots + c_{k-1}T(v_{k-1}) + c_kT(v_k). \end{aligned}$$

T is linear iff T takes linear combinations to linear combinations.

Example: Suppose $T(v_1) = 0 = T(v_2)$. Then T takes $\text{span}\{v_1, v_2\}$ to

$$T(c_1v_1 + c_2v_2) = c_1T(v_1) + c_2T(v_2) = 0 + 0 = 0.$$

Theorem 3: Any linear map takes the zero vector to the zero vector.

Proof: Since $0 + y = y$,

$$T(0 + y) = T(0) + T(y) \Rightarrow T(y) = T(0) + T(y) \Rightarrow T(0) = 0.$$

Example: Let $T(x) = Ax$ where $A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \\ 1 & 1 \end{bmatrix}$ and $b = \begin{bmatrix} 7 \\ 5 \\ 7 \end{bmatrix}$.

Domain of T is $\mathbb{R}^2$ (2 columns), Codomain of T is $\mathbb{R}^3$ (3 rows).

Is b in the image of T ? Solve $Ax = T(x) = b$:

$$\left[\begin{array}{cc|c} 1 & 1 & 7 \\ 0 & 1 & 5 \\ 1 & 1 & 7 \end{array} \right] \rightarrow \left[\begin{array}{cc|c} 1 & 1 & 7 \\ 0 & 1 & 5 \\ 0 & 0 & 0 \end{array} \right] \rightarrow \left[\begin{array}{cc|c} 1 & 0 & 2 \\ 0 & 1 & 5 \\ 0 & 0 & 0 \end{array} \right]$$

Yes, we have a consistent solution: $x = \begin{bmatrix} 2 \\ 5 \end{bmatrix}$.

Definition: The standard basis vectors $e_1, \dots, e_n$ in $\mathbb{R}^n$ are the columns of the identity $n \times n$ matrix

$$I_n = \begin{bmatrix} 1 & & 0 \\ & \ddots & \\ 0 & & 1 \end{bmatrix}, \quad \text{namely } e_i = \text{column } i \text{ of } I_n.$$

Property:

$$\begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} = x_1 e_1 + \dots + x_n e_n = I_n \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}.$$

If $A = [v_1 \ \dots \ v_n]$ is an $m \times n$ matrix and $v_i = \text{column } i \text{ of } A$, then

$$Ae_i = v_i, \quad T(x) = Ax, \quad T(e_i) = v_i.$$

$$[v_1 \ \dots \ v_n] \begin{bmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix} = 1 \cdot v_1 + 0 \cdot v_2 + \dots + 0 \cdot v_n = v_1.$$

Theorem 4 (Standard matrix of a linear map): If $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is a linear map and $A = [T(e_1) \ \cdots \ T(e_n)]$ is the $m \times n$ matrix whose i th column is $T(e_i)$, then

$$T(x) = Ax, \quad \forall x.$$

Proof:

$$\begin{aligned} T(x) &= T(x_1 e_1 + \cdots + x_n e_n) \\ &\stackrel{\text{linear}}{=} x_1 T(e_1) + \cdots + x_n T(e_n) \\ &= [T(e_1) \ \cdots \ T(e_n)] \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} = Ax. \end{aligned}$$

A linear map $\mathbb{R}^n \rightarrow \mathbb{R}^m$ is another description of an $m \times n$ matrix.

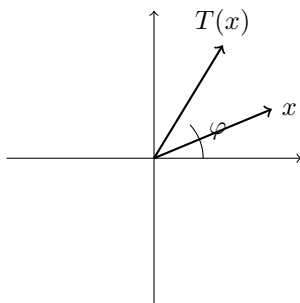
How to check if a map $\mathbb{R}^n \rightarrow \mathbb{R}^m$ is linear: Check if $T(x) = Ax$ for some A .

Example: Is $T \left(\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \right) = \begin{bmatrix} x_1 + x_2 \\ x_2 + 1 \end{bmatrix}$ linear?

$$[T(e_1) \ T(e_2)] \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} x_1 + x_2 \\ x_1 + 2x_2 \end{bmatrix} \neq \begin{bmatrix} x_1 + x_2 \\ x_2 + 1 \end{bmatrix}.$$

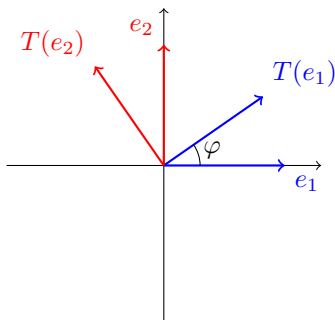
Indeed $T(0) = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \neq 0$, so by Theorem 3, T is not linear.

Example: Let $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ rotate counterclockwise by angle $\varphi \in (0, \frac{\pi}{2})$.



What's the standard matrix of T ? Since $e_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ rotates to angle φ and $e_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$ rotates to angle $\varphi + \frac{\pi}{2}$:

$$T(e_1) = \begin{bmatrix} \cos \varphi \\ \sin \varphi \end{bmatrix}, \quad T(e_2) = \begin{bmatrix} -\sin \varphi \\ \cos \varphi \end{bmatrix}.$$



So the standard matrix is

$$[T(e_1) \ T(e_2)] = \begin{bmatrix} \cos \varphi & -\sin \varphi \\ \sin \varphi & \cos \varphi \end{bmatrix}.$$

One can show that this matrix works for any φ (not just $\varphi \in (0, \pi/2)$).

Definition: A function f is **onto** if the image of f equals the codomain of f . Thus $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is onto if for every b in $\mathbb{R}^m$ there is x in $\mathbb{R}^n$ with $T(x) = b$.

Theorem 5: If $T(x) = Ax$, then T is onto $\iff$ each row of A is pivotal.

Proof: Ax is consistent for every $b \iff$ each row of A is pivotal.

(Remark: for a square matrix, invertible $\iff$ onto.)

Definition: A function f is **one-to-one** if $u \neq v \Rightarrow f(u) \neq f(v)$ for any u, v in the domain of f .

Example: If $f(x) = x^2$, $f : \mathbb{R} \rightarrow \mathbb{R}$, is f one-to-one? No.

Theorem 6: Let $T(x) = Ax$ where A is an $m \times n$ matrix. Then the following are equivalent:

1. T is one-to-one.
2. $Ax = b$ has at most one solution.
3. $Ax = 0$ only has the zero solution.
4. Each column of A is pivotal.

Proof: $2 \iff 3 \iff 4$ (given previously). $1 \iff 2$: since $T(x) = Ax$, (1) means that $T(x) = b$ has at most one solution, which is exactly what one-to-one means.

Example: Let $T\left(\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}\right) = \begin{bmatrix} 3x_1 + x_2 \\ 5x_1 + 7x_2 \\ x_1 + 3x_2 \end{bmatrix}$.

Domain of $T = \mathbb{R}^2$, Codomain of $T = \mathbb{R}^3$.

Is T linear? If so, find its standard matrix.

$$[T(e_1) \ T(e_2)] \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 3 & 1 \\ 5 & 7 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 3x_1 + x_2 \\ 5x_1 + 7x_2 \\ x_1 + 3x_2 \end{bmatrix}$$

$\therefore T$ is linear, with standard matrix $\begin{bmatrix} 3 & 1 \\ 5 & 7 \\ 1 & 3 \end{bmatrix}$.

Is T one-to-one? Onto?

$$\begin{bmatrix} 3 & 1 \\ 5 & 7 \\ 1 & 3 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 3 \\ 5 & 7 \\ 3 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 3 \\ 0 & -8 \\ 0 & -8 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix} \text{ (RREF)}$$

Both columns are pivotal, so T is one-to-one. Only 2 of the 3 rows are pivotal (row 3 is zero), so T is **not** onto.